

Support Flow Tensor Field for Fast Build-Orientation Candidate Generation in Support-Requiring Additive Manufacturing

Abstract

Build orientation strongly controls support volume in additive manufacturing, but exhaustive orientation sweeps remain costly. The proposed method targets support-requiring additive-manufacturing processes in which build orientation changes the amount of removable support material; it is not intended for inherently support-free processes. This paper proposes the Support Flow Tensor Field (SFTF), a fast candidate generator that ranks promising build directions before local verification by a support-volume grid or slicer. For each sampled direction, SFTF casts rays from overhang faces to actual receiver faces and forms a direction-dependent asymmetric flow tensor. The build-plate support term is further interpreted as a virtual ground receiver, making the dominant predictive term a unified Rayleigh cost. On five benchmark meshes with a 1° tomography grid, verifying the $\pm 10^\circ$ neighborhoods of the top-20 SFTF candidates uses 3.57% of the grid on average and obtains a mean support-volume ratio of 1.13 with maximum 1.58. Budget-matched uniform and random local-window controls are weaker, with mean/maximum ratios of 1.64/2.79 and 2.62/5.44. The paper also introduces Adaptive Verification Escalation (AVE), an automatic hard-case escalation rule based on local-verification confidence signals. On the five stored 1° meshes it reduces the worst ratio from 8.52 to 1.06 while using 2.99% of the full grid on average; on a five-group cached 1° audit, it reduces the maximum ratio from 17.57 to 1.37 at a mean budget of 4.55%. These results show that SFTF is best viewed not as a replacement for precise support-volume optimization, but as a low-cost warm start that concentrates expensive verification on a small number of likely basins.

Keywords: Support Flow Tensor Field (SFTF), build orientation, support volume, support-requiring additive manufacturing, local verification, candidate generation

1 Introduction

Build orientation strongly affects support volume, material use, print time, and post-processing quality in support-requiring additive manufacturing, especially FDM processes in which downward overhangs beyond a critical angle require removable support [5]. Existing explicit or voxel/slicer-based orientation searches can evaluate support volume accurately, but exhaustive yaw-pitch sweeps remain expensive because they evaluate a dense set of directions even for simple shapes.

1.1 Related Work

Build-orientation studies include explicit support-volume evaluation, voxel/depth-normal acceleration, support-structure design, and topology or multi-axis design approaches [2, 3, 6, 10, 12–14]. Shadow-volume and tomographic formulations estimate support volume without full slicer generation and can also be accelerated on GPUs [4, 8, 9]. In this manuscript, TOMO_CPU and TOMO_CUDA denote the CPU and GPU tomographic support-volume evaluators used as verification and comparison backends [9]; SFTF is instead a candidate generator that reduces the number of directions passed to such verifiers.

Tensor or anisotropic formulations have also appeared in topology optimization: overhang constraints, anisotropic interface energy, and deposition-direction-dependent stiffness can bias a redesigned part toward self-supporting geometry [? ? ?]. The present work is complementary: the input shape is fixed, and the tensor is used only to rank build orientations before an accurate TOMO or slicer verification step. A related industrial patent seeks build orientations for fixed complex parts but does not disclose an open algorithmic procedure [?].

1.2 Contribution

This paper proposes the Support Flow Tensor Field (SFTF), a low-cost candidate generator rather than a replacement for precise TOMO/slicer optimization. For each sampled build direction, SFTF estimates internal face-to-face support flow by ray casting, adds a build-plate contribution through a virtual ground node, and then passes only a small set of promising basins to local TOMO verification. The main contributions are: (i) a direction-dependent asymmetric support-flow tensor for fixed parts; (ii) a ground-node derivation showing that the build-plate term is an exact Rayleigh contribution of the augmented tensor; (iii) a local-verification budget curve demonstrating that SFTF top-20 windows recover a mean support-volume ratio of 1.13 with only 3.57% of the 1° grid; (iv) Adaptive Verification Escalation (AVE), an automatic escalation rule for narrow-basin hard cases; and (v) a five-group timing audit showing $8.3\text{--}381.9\times$ Python and $56.8\text{--}56,027\times$ C++ speedups relative to TOMO_CPU on the same 1° sweep.

2 Method

2.1 From Symmetric Support Tensor to Directional Flow

As a symmetric baseline, we define an area-weighted second-moment tensor of face normals

$$M = \sum_i A_i m_i m_i^T, \quad (1)$$

following the general orientation/fabric-tensor construction for directional data [?]. Here A_i and m_i are the area and outward normal of triangular face i . In this paper, this tensor is used only as a support-oriented baseline: it summarizes the distribution of surface normals, but it does not encode whether a face is an overhang for a given build direction, whether another face can actually receive the support, or how far an unsupported face is from the build plate. SFTF therefore evaluates a direction-dependent asymmetric tensor for each unit build direction $n \in S^2$.

For face centroid c_i , the overhang coefficient is

$$O_i(n) = \max(0, -m_i \cdot n). \quad (2)$$

For sampled overhang source faces, a ray is cast along $-n$; if the first valid receiver is face j , the height is $h_{ij}(n) = (c_i - c_j) \cdot n$ and the face-to-face influence is

$$w_{ij}(n) = \frac{O_i(n)}{1 + \alpha h_{ij}(n)}, \quad \alpha = 1/D, \quad (3)$$

where D is the mesh bounding-box diagonal. The face-to-face support flow tensor is

$$F(n) = \sum_{(i,j) \in \mathcal{R}(n)} w_{ij}(n) A_i A_j m_i m_j^T, \quad (4)$$

with $\mathcal{R}(n)$ denoting valid ray-cast support relationships. The receiver must not be the source face, must have positive height, and must face the build direction sufficiently ($m_j \cdot n > 0.05$). To bound cost, source faces are sampled deterministically in proportion to $A_i O_i(n)$ using

$$K_{ray} = \text{clip}(0.02 N_f, 500, 3000), \quad (5)$$

where N_f is the number of mesh faces.

2.2 Ground-Node Build-Plate Term

If no valid receiver face is found, the overhang is ultimately supported from the build plate. With

$$z_{plate}(n) = \min_{v \in V} v \cdot n, \quad h_{iP}(n) = c_i \cdot n - z_{plate}(n), \quad (6)$$

the scalar plate penalty is

$$B(n) = \sum_{i \in \mathcal{B}(n)} O_i(n) A_i (1 + \alpha h_{iP}(n)). \quad (7)$$

This term is not ad hoc: as illustrated in Fig. 1, it is exactly the Rayleigh contribution obtained by adding a virtual receiver whose normal is the build direction n ,

$$\tilde{F}(n) = \sum_{(i,j) \in \mathcal{R}(n)} w_{ij}(n) A_i A_j m_i m_j^T + \sum_{i \in \mathcal{B}(n)} A_i (1 + \alpha h_{iP}) m_i n^T. \quad (8)$$

Since $m_i \cdot n = -O_i(n)$, the symmetric Rayleigh contraction of the ground term contributes exactly $B(n)$ to the support cost, giving

$$\tilde{R}(n) = \max(0, -n^T \tilde{F}_s(n) n) = R(n) + B(n). \quad (9)$$

The equality was numerically confirmed within 10^{-12} over the stored candidate pools.

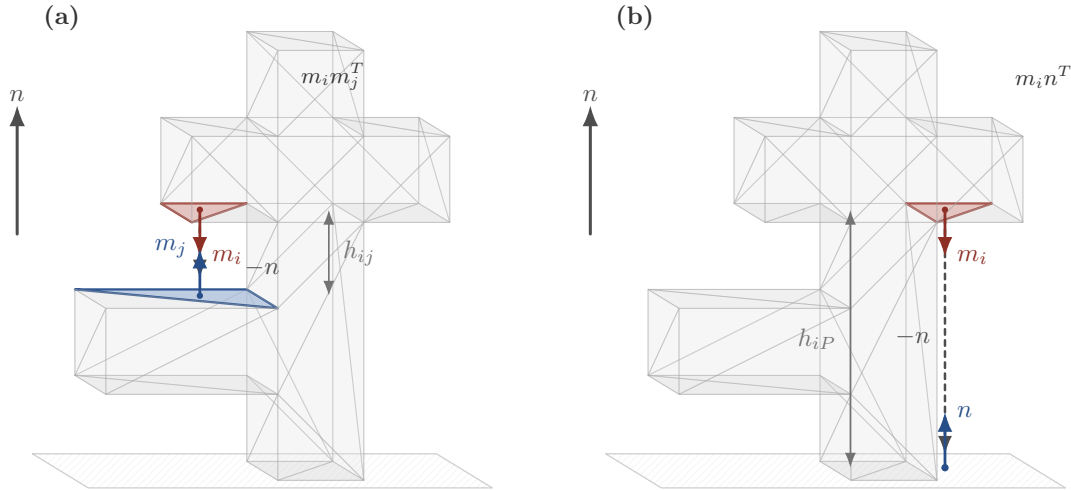


Fig. 1 Unified flow interpretation. Face-to-face support uses receiver normal m_j , while direct build-plate support is represented by a virtual receiver with normal n ; the latter gives the Rayleigh contribution $B(n)$ in Eq. (9).

2.3 Candidate Generation and Rescoring

The deployed score is

$$J(n) = \tilde{R}(n) + P(n), \quad P(n) = \sum_{(i,j) \in \mathcal{R}(n)} O_i(n) A_i A_j, \quad (10)$$

where P is the face-to-face overhang amount after the height attenuation in w_{ij} cancels. A 512-direction Fibonacci-sphere sample is first evaluated, the best directions are locally resampled in 2° – 8° cones, and angular non-maximum suppression keeps separated candidates. To avoid treating dimensionally different quantities as directly commensurate, the final candidate order

uses rank-normalized features¹ $[J, R, P, B, H, S, \sigma_1]$, following the standard rank-transformation idea for scale-robust empirical modeling [?]. Here H is the valid-hit count and S, σ_1 are singular-value features retained only for post-processing. The reported calibration uses weights $(w_J, w_R, w_P, w_B, w_H, w_S, w_1) = (0, 0.731, -2.3897, 2.1447, 2.3109, -2.2922, 1.6363)$, and cross-validation below reports the expected loss when these weights are not fit in sample. Implementation optimizations, including first-hit fallback and per-mesh cached geometry, reduced a 1k-direction evaluation of *Bunny 69k* from about 15.7 s to 1.5 s without changing selected directions.

3 Evaluation Protocol and Datasets

3.1 Datasets and Experimental Environment

The dataset consists of five groups that deliberately separate and control the two axes of *shape difficulty* and *mesh scale*. A–C set the scale in the $\sim 10^1$ – 10^5 -face range and vary the difficulty axis from basic shapes to freeform organic shapes, revealing the *fixed-cost floor* of the exhaustive sweep (a cost almost independent of mesh complexity). D–E fix the difficulty as real manufacturing parts while pushing the scale axis from 256 faces to 1.04×10^6 faces, revealing the *adaptive cost* by which the SFTF candidate-evaluation cost increases adaptively to mesh complexity. Therefore A–E is not a single size ramp, and the partial overlap of the face-count ranges of groups C and D is the result of a design that separates and controls the two axes, not an ordering error.

- **A (basic shapes, 5; lower difficulty):** cube, sphere, cylinder, cone, torus (face count $\sim 10^1$ – 10^3).
- **B (simple functional parts, 5; mid difficulty):** u_bracket, hook, c_clamp, pipe_elbow, hollow_box.
- **C (organic/freeform benchmarks, 5; upper difficulty):** standard scan/sculpture meshes such as *Bunny 69k*, *Manikin 13k*, *Dragon 100k*, *Happy 50k*, and *Lucy 50k*. The Stanford grid meshes used in §4.3 are exactly this group.
- **D (Thingi10K set 1, 10; lower scale):** real manufacturing parts with 256–88,106 faces [?].
- **E (Thingi10K set 2, 10; upper scale):** large high-resolution meshes with about 5.1×10^5 – 1.04×10^6 faces.

All timing experiments were run on a Windows 11 Education 64-bit workstation with an AMD Ryzen 9 9950X3D CPU (16 cores, 32 logical threads), 125 GB RAM, and an NVIDIA GeForce RTX 5080 GPU (16 GB, driver 595.79). The Python implementation used Python 3.12.13 with NumPy 1.26.4, SciPy 1.17.1, Trimesh 4.4.0, Matplotlib 3.11.0, and scikit-learn 1.9.0. TOMO_CPU and TOMO_CUDA were called through Windows DLL interfaces, and the SFTF_C++ implementation was built as a Windows x64 C++ DLL using the Visual Studio 2022/MSVC toolchain with OpenMP parallelization.

3.2 Comparison Targets and Evaluation Protocol

We first define the comparison targets and evaluation procedure used in the quantitative analyses up to §4.4. The comparison targets are as follows.

¹In this paper, rank tuning denotes fitting weights after converting each feature to its within-candidate rank; it is an empirical candidate-ranking calibration, not continuous geometric optimization.

1. PCA-based direction
2. Existing Support Tensor eigenvector direction
3. Initial eigenvector-based concave direction
4. This study’s Support Flow Tensor Field direction
5. TOMO_CPU or commercial-slicer-based v_{ss} optimal direction

For each mesh, we generate a v_{ss} contour at 1° intervals under a critical angle of 60° , and evaluate how close the v_{ss} value at our method’s top candidate directions is to the global optimum. Even if the direction coordinates do not match exactly, if the v_{ss} values are similar we regard them as optimal orientations of the same level. Also, considering the case where the global optimum appears scattered across multiple basins, we compare angularly separated top- k basin representatives rather than simple top- k grid points. The candidate basins proposed by SFTF are re-input to TOMO_CPU to directly measure the Mss, and based on that measured value we mark the measured optimal and measured worst basins. Unless otherwise stated, both the five-mesh accuracy results and the A–E implementation audit use stored 1° TOMO_CPU grids. Ratio statistics at $\theta_c = 60^\circ$ exclude records whose TOMO global best is nonpositive, because those cases make a positive support-volume ratio ill-defined.

4 Results and Discussion

4.1 Stored TOMO Grid Comparison

The stored 1° , $\theta_c = 60^\circ$ TOMO_CPU grids were used as the accuracy reference. SFTF candidates were mapped to the nearest signed TOMO yaw–pitch cell, and ratios were computed against the global TOMO best value. Table 1 reports the calibration-set comparison; the rank-tuned values are intentionally treated as in-sample and are evaluated again by cross-validation below.

Table 1 Comparison of the stored TOMO_CPU v_{ss} grid with the current SFTF candidates. Ratio is the ratio of the SFTF candidate’s v_{ss} to the TOMO global optimal v_{ss} .

Mesh	TOMO best	SFTF score1	Ratio	SFTF best-of-3	Ratio
<i>Bunny 69k</i>	3201.39	4665.85	1.46	4665.85	1.46
<i>Manikin 13k</i>	4657.72	8675.96	1.86	8675.96	1.86
<i>Dragon 100k</i>	24446.98	45804.41	1.87	39547.57	1.62
<i>Happy 50k</i>	2112.47	25467.94	12.06	23727.49	11.23
<i>Lucy 50k</i>	5036.14	14192.68	2.82	14192.68	2.82

Raw ranking alone is insufficient for the *Happy 50k* hard case, so SFTF is used as a warm start for local TOMO verification rather than as a standalone optimizer.

4.2 Local Verification and Adaptive Verification Escalation

The deployed metric is a budget curve: the top- K SFTF candidates define $\pm 10^\circ$ local windows, and the best TOMO grid value inside their union is selected. This changes the interpretation from whether the top-3 ranking alone solves the problem to whether SFTF places a small verification budget near useful TOMO basins. Table 2 summarizes this budget–accuracy trade-off.

Table 2 Budget curve of top- K local TOMO verification around SFTF candidates. The budget is the mean fraction of the 1° TOMO grid covered by the union of local windows.

Top- K	Mean ratio	Median ratio	Max ratio	Mean grid budget	Ratio ≤ 2.0
3	2.59	1.15	8.52	0.87%	4/5
10	2.52	1.00	8.52	2.12%	4/5
20	1.13	1.00	1.58	3.57%	5/5
30	1.01	1.00	1.06	5.43%	5/5

Budget-matched controls in Table 3 show that the gain is not only the amount of grid inspected; SFTF places the windows better than uniform-axis or random seeds.

Table 3 Budget-matched local TOMO controls at the per-mesh SFTF top-20 grid-cell budget. SFTF improves both the mean and worst-case ratio, showing that the warm-start value is the placement of local windows, not only the budget size.

Policy	Mean	Median	Max	Budget	≤ 2.0
SFTF top-20 local	1.13	1.00	1.58	3.57%	5/5
Uniform-axis matched	1.64	1.38	2.79	3.57%	4/5
Random matched (median)	2.62	1.75	5.44	3.57%	3/5

Table 4 reports the resulting automatic escalation policy. Adaptive Verification Escalation (AVE) escalates only when ordinary top-10 local verification is uncertain. The trigger uses local-verification quantities only,

$$\begin{aligned} \text{trigger}_{\text{AVE}} &= ([\eta \geq 0.03] \wedge [\text{edge} = 1 \vee g \leq 1.5]) \vee [\text{edge} = 1 \wedge g \geq 3.0], \\ \eta &= V_{ss}^{\text{loc}} / V_{\text{bbox}}, \quad g = V_{ss}^{\text{seed}} / V_{ss}^{\text{loc}}. \end{aligned} \quad (11)$$

Here $\text{edge} = 1$ indicates that the best local cell lies within 1° of a window boundary. The absolute threshold $\eta \geq 0.03$ may also fire on genuinely support-heavy parts, but escalation only expands the verified candidate set; it can spend more budget but cannot worsen the selected verified value.

Table 4 Automatic Adaptive Verification Escalation (AVE) from top-10 local TOMO verification. The detector of Eq. (11) selected only *Happy 50k* for escalation.

Policy	Expanded	Mean	Max	Budget	≤ 2.0
Top-10 local	0/5	2.52	8.52	2.12%	4/5
AVE	1/5	1.01	1.06	2.99%	5/5
AVE+Pareto	1/5	1.01	1.06	3.18%	5/5
All-mesh top-30	5/5	1.01	1.06	5.43%	5/5

The five stored meshes are in-sample for the detector thresholds. Table 5 reports the held-out cached A–E audit, which contains 16 ratio-valid records after excluding 19 records with nonpositive TOMO best values at $\theta_c = 60^\circ$.

Table 5 Cached five-group audit of the tiered AVE policy on ratio-valid 1° , $\theta_c = 60^\circ$ TOMO_CPU records. The top-10 baseline over all ratio-valid records has mean ratio 2.72, maximum 17.57, mean budget 2.40%, and 13/16 records below ratio 2.0.

Group	n	Expanded	Mean	Max	Budget	≤ 2.0
A	1	1	1.04	1.04	6.32%	1/1
B	1	1	1.24	1.24	4.79%	1/1
C	5	1	1.01	1.06	2.99%	5/5
D	5	3	1.22	1.37	6.84%	5/5
E	4	1	1.15	1.34	3.13%	4/4
All	16	7	1.13	1.37	4.55%	16/16

The audit reduces the maximum ratio from 17.57 to 1.37 and keeps all ratio-valid records below 2.0 with a mean budget of 4.55%.

4.2.1 Representative TOMO Landscapes

Figures 2 and 3 show the contrast between a broad-basin ordinary case and the *Happy 50k* hard case. Additional *Bunny 69k*, *Manikin 13k*, and *Lucy 50k* landscapes are provided in Supplementary Figs. S5–S7.

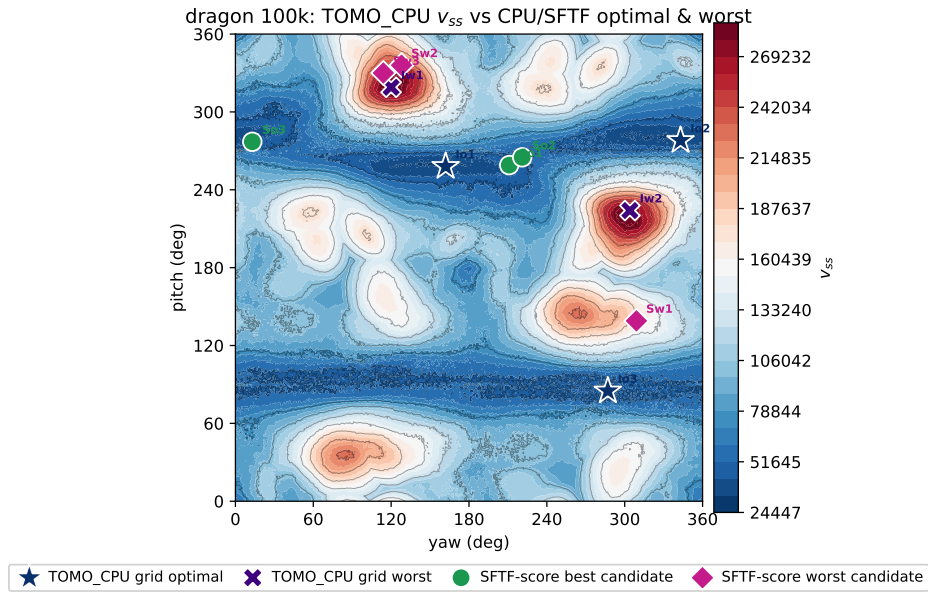


Fig. 2 Representative TOMO_CPU v_{ss} landscape of *Dragon 100k*. The global optimum forms a distinct low-cost basin, illustrating a non-hard case where coarse SFTF candidates already cover useful regions.

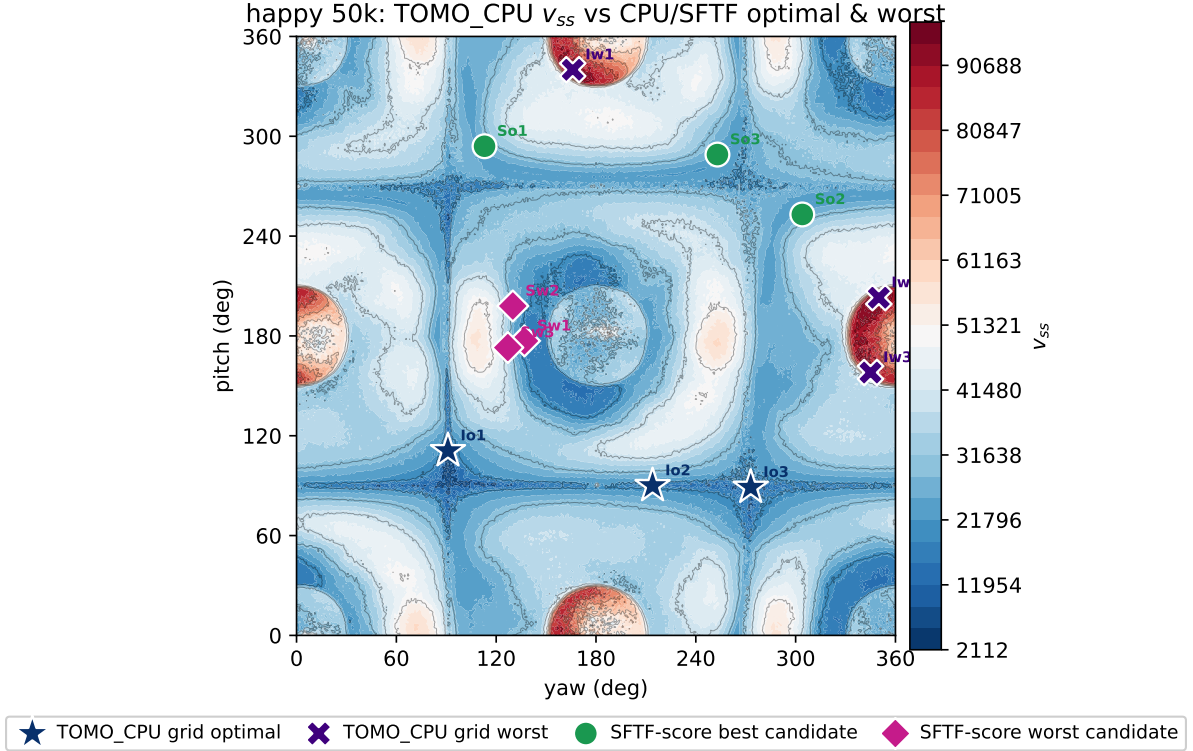


Fig. 3 *Happy 50k* before and after AVE on the same TOMO_CPU v_{ss} landscape. The base top-10 local verification selects a high-cost basin ($v_{ss} = 17,991$, ratio 8.52), whereas AVE recovers the TOMO optimum ($v_{ss} = 2,112$, ratio 1.00).

4.3 Feature Contribution and Generalization

Table 6 reports Spearman rank correlations over the full candidate pools and shows that the build-plate term is the most stable single predictor of TOMO v_{ss} .

Table 6 Spearman rank correlation between features and TOMO v_{ss} over the entire candidate pool. The larger the positive value, the more valid the feature is as a cost.

Feature	<i>Bunny 69k</i>	<i>Manikin 13k</i>	<i>Dragon 100k</i>	<i>Happy 50k</i>	<i>Lucy 50k</i>	Average
$J = R + P + B$	0.46	0.26	0.80	0.27	0.57	0.47
R (Rayleigh)	0.40	-0.33	-0.57	-0.03	0.14	-0.08
P (pair flow)	-0.04	-0.38	-0.72	0.00	-0.59	-0.34
B (build-plate)	0.50	0.66	0.85	0.29	0.62	0.58
H (hit count)	0.24	-0.53	-0.73	-0.00	-0.71	-0.35
S (nuclear)	-0.02	-0.36	-0.70	-0.05	-0.26	-0.28
σ_1	-0.03	-0.34	-0.66	-0.07	0.07	-0.21

The full objective ablation is moved to Supplementary Table S5. It shows that adding nuclear/ σ_1 terms does not alter candidate selection, while tensor-only variants are weakest. Table 7 then separates the unified Rayleigh term from its components. The ground-node reformulation explains why B is not merely a scalar patch: the unified Rayleigh term $\tilde{R} = R + B$ is a tensor contraction and has the strongest average single-feature correlation.

Table 7 Spearman correlation of the unified Rayleigh term $\tilde{R} = R + B$ and single terms with TOMO v_{ss} (higher is better). $\tilde{R}$ was computed directly from the augmented tensor and agrees with $R + B$ within 10^{-12} .

Mesh	R	B	$\tilde{R} = R+B$	$J = R+P+B$
<i>Bunny 69k</i>	0.40	0.50	0.52	0.46
<i>Manikin 13k</i>	-0.33	0.66	0.65	0.26
<i>Dragon 100k</i>	-0.57	0.85	0.89	0.80
<i>Happy 50k</i>	-0.03	0.29	0.31	0.27
<i>Lucy 50k</i>	0.14	0.62	0.62	0.57
Average	-0.08	0.58	0.60	0.47

The best-of-3 comparison in Supplementary Table S6 shows the same trend: excluding the *Happy 50k* hard case, removing the negative-signal P term improves the mean ratio from 2.82 to 2.50. A height-convention check also supported the current sign choice: amplification for tall build-plate support and attenuation for distant face-to-face support.

Cross-validation exposes the optimism of calibration-set rank tuning. Table 8 summarizes the leave-one-mesh-out (LOMO) check: training excludes the target mesh, fits the rank-normalized weights on the others, and evaluates the held-out best-of-3 ratio.

Table 8 LOMO cross-validation best-of-3 ratio of rank tuning (lower is better). The in-sample column is the calibration-set fit value reported in Table 1. The oracle is the best direction existing within the candidate pool, representing the upper bound of candidate generation.

Mesh	oracle	untuned J	in-sample (reported)	LOMO
<i>Bunny 69k</i>	1.46	2.47	1.46	2.20
<i>Manikin 13k</i>	1.13	3.99	1.86	2.62
<i>Dragon 100k</i>	1.29	2.01	1.62	1.86
<i>Lucy 50k</i>	1.02	2.82	2.82	2.82
Average	1.22	2.82	1.94	2.37

The in-sample average 1.94 becomes 2.37 under LOMO, but remains better than PCA (3.63) and the symmetric Support Tensor baseline (4.30). Table 9 summarizes this comparison. The larger five-group 1° nested CV is less favorable for rank-only scoring: the family-held-out selected model has mean ratio 11.68, median 2.21, and maximum 137.73, with the *Happy 50k* hard case and Group D3 dominating the tail. This reinforces the role of rank tuning as a candidate-ordering heuristic, not as the final optimizer; the reliable result is obtained when SFTF candidates are followed by local TOMO verification as in §4.2.

Table 9 Unified comparison of existing methods and SFTF (best-of-3 signed v_{ss} ratio, lower is better). SFTF-tuned is the five-mesh LOMO estimate for rank-only candidate ordering.

Mesh	PCA	Support Tensor	SFTF untuned	SFTF tuned (LOMO)	oracle
<i>Bunny 69k</i>	2.82	3.71	2.47	2.20	1.46
<i>Manikin 13k</i>	2.65	2.50	3.99	2.62	1.13
<i>Dragon 100k</i>	2.69	2.33	2.01	1.86	1.29
<i>Lucy 50k</i>	6.37	8.67	2.82	2.82	1.02
Average	3.63	4.30	2.82	2.37	1.22

4.4 Five-Group Implementation Audit

The A–E audit separates shape difficulty from mesh scale.

It compares TOMO_CPU, TOMO_CUDA, Python SFTF, and C++ SFTF on the same 1° , $\theta_c = 60^\circ$ sweep setting. The full per-mesh timing table is in Supplementary Table S1; Table 10 keeps the group averages.

Table 10 Group-wise average execution time and speed ratio. The C++ SFTF average is in ms, and the TOMO and Python SFTF averages are in seconds. The speed ranges use the same definitions as Supplementary Table S1: TOMO_CPU over Python SFTF and TOMO_CPU over SFTF_C++.

Group	TOMO		SFTF		comparison	
	CPU (s)	CUDA (s)	py (s)	C++ (ms)	CPU/py	CPU/C++
A	42.75	15.36	0.32	8.9	54.4–381.9×	1,683–50,641×
B	42.45	14.94	0.38	9.3	58.0–269.0×	1,857–56,027×
C	51.48	60.13	1.85	183.6	23.3–36.9×	192–854×
D	53.58	112.38	1.97	124.5	12.8–128.8×	213–8,920×
E	643.75	740.88	15.57	3123.5	8.3–59.1×	56.8–258.6×

Over all 35 meshes, Python SFTF is 8.3–381.9× faster than TOMO_CPU, while SFTF_C++ is 56.8–56,027× faster. TOMO_CUDA helps on simple A/B shapes but is often slower than CPU on organic and large meshes in this environment, so the dominant gain is algorithmic: reducing the number of directions to evaluate, not merely changing the exhaustive-sweep backend. Group E confirms bounded scaling: the clipped ray budget in Eq. (5) caps per-direction ray cost at 3000 samples, so large meshes near 10^6 faces remain feasible. Supplementary Table S2 gives the $R/P/B/\text{hit}$ decomposition, Supplementary Table S3 gives the critical-angle sweeps, Supplementary Fig. S1 shows the five-group best/worst overlay, and Supplementary Figs. S2–S4 visualize the *Bunny 69k* K_{ray} trade-off. These results support the intended role of SFTF: a fast, interpretable candidate generator whose outputs are verified locally by TOMO or a slicer.

5 Conclusion

This study proposed a direction evaluation method that combines the Support Flow Tensor Field and a build-plate penalty for build-orientation optimization in additive manufacturing. The eigenvector interpretation of the existing symmetric tensor is useful for shape directionality, but it does not sufficiently reflect direction-dependent overhangs, internal visibility, and the build-plate support requirement.

The current algorithm directly computes an asymmetric Support Flow Tensor for each candidate direction, and constructs a basic objective function $J = R + P + B$ that combines the Rayleigh support cost, the face-to-face support volume, and the build-plate penalty. In the actual final candidate selection, it applies a rank-normalized tuning score T of $J, R, P, B, H, S, \sigma_1$ to a candidate pool made by coarse spherical sampling and small-angle refined sampling, to choose promising orientations to pass to slicer/TOMO verification. When used this way, verifying the $\pm 10^\circ$ neighborhoods of only the top-20 candidates recovers a mean support-volume ratio of 1.13 (maximum 1.58) while inspecting 3.57% of the 1° grid, and the AVE automatically widens this budget only for detected hard cases (§4.2).

The quantitative analysis (§4.3) shows both the strengths and limitations of the method. On the five stored meshes, cross-validated SFTF rank tuning is superior to PCA and Support Tensor eigenvector directions, supporting the starting assumption. At the same time, the five-group 1° nested CV shows that rank-only scoring is not sufficiently reliable as a standalone optimizer. Most of the predictive power originates from the build-plate penalty and the internal

visibility evaluation, the marginal contribution of some tensor-derived terms such as $\text{nuclear}/\sigma_1$ is small, and the calibrated rank tuning weights show an optimistic bias on unseen meshes. Meanwhile, in §2.2–§4.3 we showed that the build-plate penalty is exactly equivalent to the Rayleigh contribution obtained by adding a virtual ground node to the flow tensor. This unifies the term driving the predictive power into the tensor contraction $\tilde{R} = R + B$, justifying the “tensor” formulation, and confirms that $\tilde{R}$, excluding the negative-signal face-to-face support volume P , is consistently superior to the existing J .

Therefore, in future work we will (i) determine the weights on a cross-validation basis rather than a calibration set to control overfitting, (ii) remove terms with marginal contribution or replace them with features highly correlated with the actual v_{ss} , such as face-to-face self-support coverage or a support-volume proxy, (iii) normalize the dimensions of the objective-function terms to secure the physical meaning and transferability of the weights, (iv) add projection-area, print-height, and contact-stability terms to extend to a multi-objective orientation optimization model, and (v) as observed in §4.4, since the TOMO optimal orientation shifts with the critical angle θ_c while the current SFTF candidates are invariant to θ_c , extend to θ_c -dependent candidate generation with overhang truncation turned on. For cases where the ordinary top- K candidate generation does not include a stable near-optimal basin, AVE provides an automatic escalation path, and future work will refine its thresholds on a larger family-held-out validation set.

Statements and Declarations

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Competing interests

The author declares no competing interests.

Data availability

The numerical results reported in this manuscript are based on cached mesh-evaluation records, support-volume grids, and generated summary tables maintained with the research code. An anonymized repository or supplementary archive can be supplied during peer review, and the permanent public repository link can be provided after acceptance. Third-party mesh models should be obtained from their original sources subject to their respective licenses.

Code availability

The code and scripts supporting the reported results will be made publicly available upon acceptance. An anonymized review archive can be provided during peer review if required.

Use of artificial intelligence tools

AI coding assistants, including OpenAI Codex and Anthropic Claude Code, were used to assist with portions of research-code implementation, figure-generation scripts, and manuscript-support automation. All generated code, experimental results, and manuscript changes were reviewed and validated by the author.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

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